Raphaël Terrine

3rd year PhD student at Polytechnique Institute

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Research Interests

I am interested in the study of Partial Differential Equations both numerically and theorically and the study of inverse problems using optimal control and one shot methods.
I'm also interested in the implementation and the analysis of the Finite Elements Method.

Education

2023 – 2026 **PhD in Applied Mathematics** ENSTA Paris/Polytechnique
Thesis: Identification of bottom deformations of the ocean from surface measurements.

2020 – 2023 **Engineering school** ENSTA Paris
Specialisation : Applied maths and waves propagation.

2022 – 2023 **Master's degree in applied Mathematics** Polytechnique Institute
Specialisation : Modelisation, Simulation and Analysis.

2018 – 2020 **French Preparatory Classes MPSI-MP*** Lycée Henri IV
Two-year intensive preparatory program for competitive entrance to French Grandes Écoles.

Professional experiences

Summer 2023 **Pre PhD internship in POEMS team** ENSTA Paris
Identification of bottom deformations of the ocean from surface measurements.

Summer 2022 **Research Internship supervised by G. Bal** University of Chicago
Analysis of transport along the boundaries of domains generated by Coriolis forces in geophysical fluid flows.

Publications

2024 **L. Bourgeois, J-F. Mercier, R. Terrine** Inverse Problems and Imaging
Identification of bottom deformations of the ocean from surface measurements.

In preparation **L. Bourgeois, P. Moireau, R. Terrine**
About convergence of a discretized optimal control approach: application to a tsunami inverse problem.

Talks

29 October 2025 **Invited speaker at PICOF'2025** Hamammet
Convergence of a numerical scheme for solving an inverse problem : Application to tsunami reconstruction.

5 June 2025 **SMAI 2025** Carcans
Identification of bottom deformations from surface measurements.

4 July 2024 **Waves 2024** Berlin
About the identification of tsunamis from surface measurements.

Teaching

2023-2025 **APM-3MA02-TA : L3 level class** ENSTA Paris
Outils élémentaires d'analyse pour les equations aux dérivées partielles.

Technical Skills

Programming: Python, C++, MATLAB
Tools: LaTeX, Git
Languages: French (native), English (fluent)